

1. WAFT ISO 7010 Safety Symbols Usage Guide

Complete Guide for Using ISO 7010 Symbols in WAFT
Practical Examples and Best Practices

1.1. Quick Start

1.1.1. Installation

The `typsium-iso-7010` package is automatically available in WAFT Typst documents. No installation needed - just import it:

```
#import "@preview/typsium-iso-7010:0.1.0": *
```

1.1.2. Basic Usage

```
#warning-sign(1, height: 2cm)
#fire-sign(1, height: 2cm)
#emergency-sign(1, height: 2cm)
```

1.2. Using WAFT Wrapper Functions

WAFT provides convenient Python functions for generating documents with ISO 7010 symbols.

1.2.1. Method 1: `generate_worldbuild_iso()`

Full control over symbol placement:

```
from pathlib import Path
from src.waft.templates.typst.wrappers.worldbuild_iso import generate_worldbuild_iso

safety_symbols = [
    {
        "function": "warning-sign",
        "code": 3, # Electric shock
        "label": "Electrical Hazard",
        "description": "High voltage equipment. Do not touch."
    },
    {
        "function": "fire-sign",
        "code": 1, # Fire extinguisher
        "label": "Fire Extinguisher",
        "description": "Located 10 meters to your right."
    }
]

pdf_path = generate_worldbuild_iso(
    title="Safety Information",
    content="Your document content...",
    output_path=Path("output/safety_info.pdf"),
    doc_id="SAFETY-001",
    safety_symbols=safety_symbols
)
```

1.2.2. Method 2: `generate_worldbuild_with_symbols()`

Simpler method using predefined symbol names:

WAFT-ISO-7010	WAFT ISO 7010 Usage Guide	2 / 8
<pre>from src.waft.templates.typst.wrappers.worldbuild_iso import generate_worldbuild_with_symbols pdf_path = generate_worldbuild_with_symbols(title="Laboratory Safety Guide", content="Your content...", output_path=Path("output/lab_safety.pdf"), symbols=["warning", "electric", "fire", "emergency", "first_aid"])</pre> Available predefined symbols: • "warning" - General warning • "danger" - Danger • "electric" - Electric shock • "radiation" - Ionizing radiation • "biohazard" - Biological hazard • "fire" - Fire extinguisher • "emergency" - Emergency exit • "first_aid" - First aid 1.3. Direct Typst Usage For custom Typst templates, use symbols directly in your .typ files. 1.3.1. Import Statement Always import at the top of your Typst file: <pre>#import "@preview/typsium-iso-7010:0.1.0": *</pre> 1.3.2. Basic Functions <pre>// Warning signs (yellow triangle) #warning-sign(n, height: <size>) // Fire signs (red square) #fire-sign(n, height: <size>) // Emergency signs (green square) #emergency-sign(n, height: <size>) // Directional arrows #emergency-arrow(direction: "up" "down" "left" "right", height: <size>) #fire-arrow(direction: "up" "down" "left" "right", height: <size>)</pre> 1.4. Common Usage Patterns 1.4.1. Inline with Text <pre>#warning-sign(3, height: 1em)</pre> indicates electrical hazards. Example:  indicates electrical hazards.		
WAFT-ISO-7010	WAFT Project Documentation	2 / 8

1.4.2. In Grids

```
#grid(  
  columns: 3,  
  column-gutter: 12pt,  
  align: center,  
) [  
  #warning-sign(1, height: 2cm),  
  #fire-sign(1, height: 2cm),  
  #emergency-sign(1, height: 2cm),  
]
```



,



,



,

1.4.3. With Labels

```
#grid(  
  columns: (1fr, 2fr),  
  align: center,  
) [  
  #emergency-sign(1, height: 2cm),  
  [  
    *Emergency Exit*  
    #linebreak()  
    Use this door in case of emergency  
  ],  
]
```



, [**Emergency Exit**
Use this door in case of emergency],

1.4.4. With Directional Arrows

```
#grid(  
  columns: 2,
```

```
align: center,  
)[  
  #emergency-sign(1, height: 2cm),  
  #emergency-arrow(direction: "right", height: 2cm),  
]
```



,



,

1.5. Symbol Reference

1.5.1. Warning Signs (Yellow Triangle)



[Warning Sign 1 (W001)
General warning

General hazard warning. Use when specific nature not covered by other signs.]



[Warning Sign 3 (W012)
Electricity

Risk of electric shock. Use near electrical equipment or high-voltage areas.]

1.5.2. Fire Signs (Red Square)



[Fire Sign 1 (F001)
Fire extinguisher

Indicates location of fire extinguisher. Red square with white symbol.]



[**Fire Sign 2 (F002)**
Fire hose reel

Indicates location of fire hose reel. Use to mark fire hose storage.]

1.5.3. Emergency Signs (Green Square)



[**Emergency Sign 1 (E001/E002)**
Emergency exit

Indicates emergency exit or escape route. Green square with white symbol.]



[**Emergency Sign 2 (E003)**
First aid

Indicates location of first aid equipment or station.]

1.6. Best Practices

1.6.1. Symbol Sizing

Use appropriate sizes for context:

Size Recommendations:

- Inline text: height: 1em (matches text size)
- Small displays: height: 1cm to 2cm
- Standard signs: height: 3cm to 5cm
- Large displays: height: 10cm or larger

1.6.2. Consistency

Use consistent sizes for the same type of symbol throughout your document:

WAFT-ISO-7010	WAFT ISO 7010 Usage Guide	6 / 8
<pre>// Good: Consistent sizing #warning-sign(1, height: 2cm) #warning-sign(3, height: 2cm) #warning-sign(5, height: 2cm)</pre> 1.6.3. Context and Labels Always provide context or labels when the meaning might not be immediately clear: <pre>// Good: With label #grid(columns: (1fr, 2fr), align: center,)[#emergency-sign(1, height: 2cm), [*Emergency Exit* - Use in case of emergency],]</pre> 1.6.4. Integration with WAFT In WAFT projects, consider: • Adding symbols to document headers/footers • Creating dedicated safety sections • Including symbols in procedural documents • Adding to facility maps and layouts 1.7. WAFT Integration Examples 1.7.1. Example 1: Safety Information Panel <pre>from pathlib import Path from src.waft.templates.typst.wrappers.worldbuild_iso import generate_worldbuild_iso safety_symbols = [{ "function": "warning-sign", "code": 3, "label": "Electrical Hazard", "description": "High voltage equipment. Do not touch." }, { "function": "fire-sign", "code": 1, "label": "Fire Extinguisher", "description": "Located 10 meters to your right." }, { "function": "emergency-sign", "code": 1, "label": "Emergency Exit", "description": "Use in case of emergency." }] generate_worldbuild_iso(title="Facility Safety Information", content="Your content here...", output_path=Path("output/safety_panel.pdf"),</pre>		
WAFT-ISO-7010	WAFT Project Documentation	6 / 8

```
    doc_id="SAFETY-001",
    safety_symbols=safety_symbols
)
```

1.7.2. Example 2: Custom Typst Template

You can use symbols in any custom Typst template:

```
#import "@preview/typsium-iso-7010:0.1.0": *
```

```
= Laboratory Safety Guide
```

```
== Hazard Identification
```

```
#grid(
  columns: 2,
  column-gutter: 16pt,
  row-gutter: 12pt,
)[
  #warning-sign(3, height: 3cm),
  [
    *Electrical Hazard*
    #linebreak()
    High voltage equipment. Do not touch.
  ],
  #fire-sign(1, height: 3cm),
  [
    *Fire Extinguisher*
    #linebreak()
    Located at each exit.
  ],
]
```

1.8. Troubleshooting

1.8.1. Symbol Not Displaying

Problem: Symbol doesn't appear in compiled PDF.

Solutions:

1. Check import statement is at top: `#import "@preview/typsium-iso-7010:0.1.0": *`
2. Verify symbol number exists (try `#warning-sign(1, height: 2cm)` first)
3. Check for syntax errors (missing commas, etc.)

1.8.2. Wrong Symbol Appearing

Problem: Symbol number doesn't match expected ISO code.

Solution: Package uses numeric indices (1, 2, 3...) not ISO codes (W001, F001, etc.). Test different numbers to find the right symbol.

1.8.3. Size Issues

Problem: Symbol too large or too small.

Solution: Adjust the height parameter:

- Too small? Increase: `height: 5cm`
- Too large? Decrease: `height: 1cm`
- Inline? Match text: `height: 1em`

1.9. Quick Reference

1.9.1. Import

```
#import "@preview/typsium-iso-7010:0.1.0": *
```

1.9.2. Warning Signs

```
#warning-sign(1, height: 2cm) // General warning
#warning-sign(3, height: 2cm) // Electric shock
#warning-sign(5, height: 2cm) // Biological hazard
```

1.9.3. Fire Signs

```
#fire-sign(1, height: 2cm) // Fire extinguisher
#fire-sign(2, height: 2cm) // Fire hose reel
```

1.9.4. Emergency Signs

```
#emergency-sign(1, height: 2cm) // Emergency exit
#emergency-sign(2, height: 2cm) // First aid
#emergency-sign(3, height: 2cm) // Emergency telephone
```

1.9.5. Arrows

```
#emergency-arrow(direction: "right", height: 2cm)
#fire-arrow(direction: "up", height: 2cm)
```

1.10. Additional Resources

Reference Materials

- WAFT Functions: src/waft/templates/typst/wrappers/worldbuild_iso.py
- Symbol Guide: NARRATIVE-WAFT/ISO_7010_SYMBOLS_GUIDE.pdf
- Markdown Guide: docs/WAFT_ISO_7010_USAGE_GUIDE.md
- Package: @preview/typsium-iso-7010:0.1.0
- Standard: ISO 7010:2019
- Wikipedia: https://en.wikipedia.org/wiki/ISO_7010